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Using Logic in Writing

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Understanding how to create logical syllogisms does not automatically mean that writers understand how to use logic to build an argument. Crafting a logical sequence into a written argument can be a very difficult task. Don't assume that an

audience will easily follow the logic that seems clear to you. When converting logical syllogisms into written arguments, remember to:

- lay out each premise clearly
- provide evidence for each premise
- draw a clear connection to the conclusion

Say a writer was crafting an editorial to argue against using taxpayer dollars for the construction of a new stadium in the town of Mill Creek. The author's logic may look like this:

Premise 1: Projects funded by taxpayer dollars should benefit a majority of the public.

Premise 2: The proposed stadium construction benefits very few members of the public.

Conclusion: Therefore, the stadium construction should not be funded by taxpayer dollars.

This is a logical conclusion, but without elaboration it may not persuade the writer's opposition, or even people on the fence. Therefore, the writer will want to expand her argument like this:

Historically, Mill Creek has only funded public projects that benefit the population as a whole. Recent initiatives to build a light rail system and a new courthouse were approved because of their importance to the city. Last election, Mayor West reaffirmed this commitment in his inauguration speech by promising "I am determined to return public funds to the public." This is a sound commitment and a worthy pledge.

However, the new initiative to construct a stadium for the local baseball team, the Bears, does not follow this commitment. While baseball is an enjoyable

pastime, it does not receive enough public support to justify spending \$210 million in public funds for an improved stadium. Attendance in the past five years has been declining, and last year only an average of 400 people attended each home game, meaning that less than 1% of the population attends the stadium. The Bears have a dismal record at 0-43 which generates little public interest in the team.

The population of Mill Creek is plagued by many problems that affect the majority of the public, including its decrepit high school and decaying water filtration system. Based on declining attendance and interest, a new Bears stadium is not one of those needs, so the project should not be publicly funded. Funding this project would violate the mayor's commitment to use public money for the public.

Notice that the piece uses each paragraph to focus on one premise of the syllogism (this is not a hard and fast rule, especially since complex arguments require far more than three premises and paragraphs to develop). Concrete evidence for both premises is provided. The conclusion is specifically stated as following from those premises.

Consider this example, where a writer wants to argue that the state minimum wage should be increased. The writer does not follow the guidelines above when making his argument.

It is obvious to anyone thinking logically that minimum wage should be increased. The current minimum wage is an insult and is unfair to the people who receive it. The fact that the last proposed minimum wage increase was denied is proof that the government of this state is crooked and corrupt. The only way for them to prove otherwise is to raise minimum wage immediately.

The paragraph does not build a logical argument for several reasons. First, it assumes that anyone thinking logically will already agree with the author, which is clearly untrue. If that were the case, the minimum wage increase would have already occurred. Secondly, the argument does not follow a logical structure. There is no development of premises which lead to a conclusion. Thirdly, the author provides no evidence for the claims made.

In order to develop a logical argument, the author first needs to determine the logic behind his own argument. It is likely that the writer did not consider this before writing, which demonstrates that arguments which could be logical are not automatically logical. They must be made logical by careful arrangement.

The writer could choose several different logical approaches to defend this point, such as a syllogism like this:

Premise 1: Minimum wage should match the cost of living in society.

Premise 2: The current minimum wage does not match the cost of living in society.

Conclusion: Therefore, minimum wage should be increased.

Once the syllogism has been determined, the author needs to elaborate each step in writing that provides evidence for the premises:

The purpose of minimum wage is to ensure that workers can provide basic amenities to themselves and their families. A report in the Journal of Economic Studies indicated that workers cannot live above the poverty line when minimum wage is not proportionate with the cost of living. It is beneficial to society and individuals for a minimum wage to match living costs.

Unfortunately, our state's minimum wage no longer reflects an increasing cost of living. When the minimum wage was last set at \$5.85, the yearly salary of \$12,168 guaranteed by this wage was already below the poverty line. Years

later, after inflation has consistently raised the cost of living, workers earning minimum wage must struggle to support a family, often taking 2 or 3 jobs just to make ends meet. 35% of our state's poor population is made up of people with full time minimum wage jobs.

In order to remedy this problem and support the workers of this state, minimum wage must be increased. A modest increase could help alleviate the burden placed on the many residents who work too hard for too little just to make ends meet.

This piece explicitly states each logical premise in order, allowing them to build to their conclusion. Evidence is provided for each premise, and the conclusion is closely related to the premises and evidence. Notice, however, that even though this argument is logical, it is not irrefutable. An opponent with a different perspective and logical premises could challenge this argument. See the next section for more information on this issue.